



# Customer Shopping Behavior Analysis

This project analyzes transactional data from 3,900 purchases to uncover spending patterns, product preferences, subscription behavior, and customer segments. Insights will guide strategic decisions across pricing, marketing, loyalty, and product positioning.



# Project Overview & Dataset at a Glance

Dataset: 3,900 rows × 18 columns. Key features include customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color), and behavior signals (discount applied, promo code used, previous purchases, frequency, review rating, shipping type). Review Rating has 37 missing values.



# Data Preparation & Cleaning (Python)

We loaded the data with pandas, inspected structure with `df.info()` and summary stats with `.describe()`. Missing Review Rating values were imputed using the median rating per product category. Columns were standardized to snake\_case for readability and documentation.

## Feature Engineering

Added age\_group bins and purchase\_frequency\_days to capture temporal behavior.

## Redundancy Check

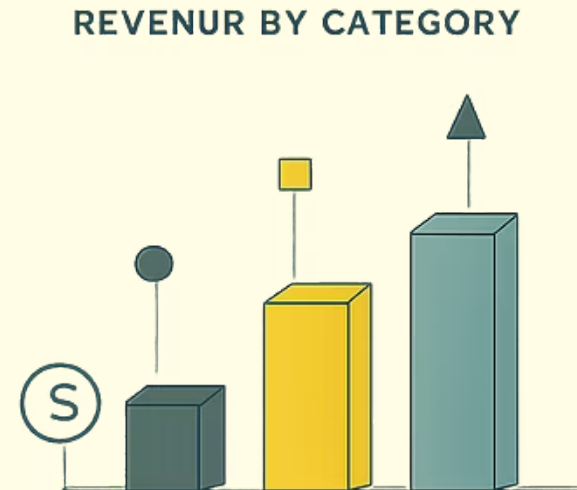
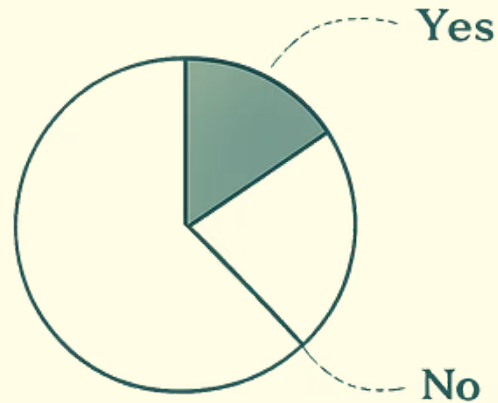
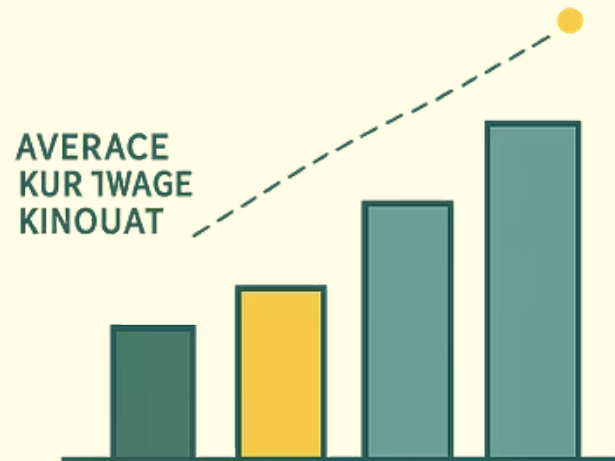
Determined discount\_applied and promo\_code\_used overlap; promo\_code\_used was dropped.

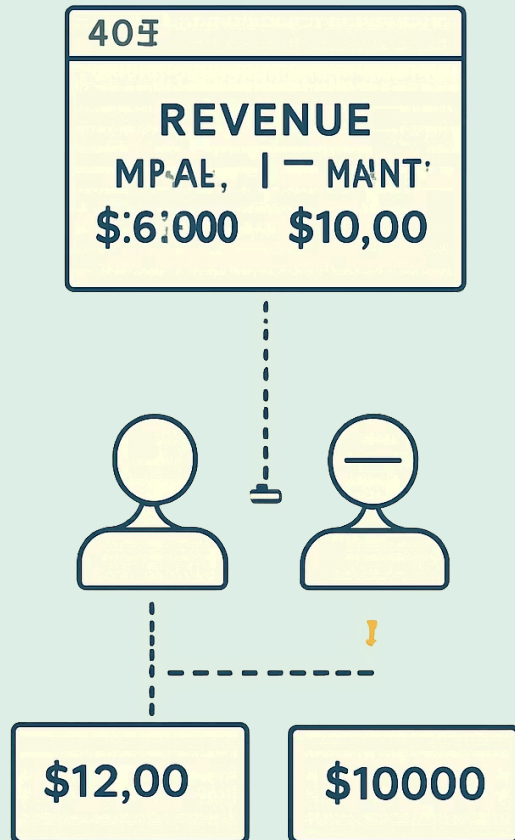
## Database Integration

Cleaned DataFrame loaded to PostgreSQL for SQL analysis and business queries.

# Key Summary Statistics

Sample descriptive stats: average purchase amount \$59.76, average review rating ~3.75, customer count 3.9K. Age mean ~44 years (std ~15), most frequent size M, top item frequency: Blouse (171). There are 50 unique locations and varied seasonality across purchases.

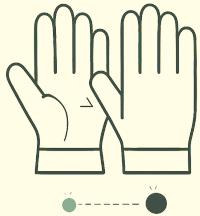




## SQL Findings – Revenue & Segments

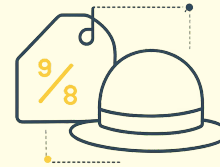
Revenue by gender: Male \$157,890 vs Female \$75,191. Customer segmentation: Loyal 3,116, Returning 701, New 83. Subscribers: 1,053 customers; Non-subscribers: 2,847. Average spend is similar across subscription status (Yes \$59.49, No \$59.87) but subscribers contribute meaningful recurring revenue.

# Top Products & Discount Patterns



## Top Rated Items

Top 5 by average rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).

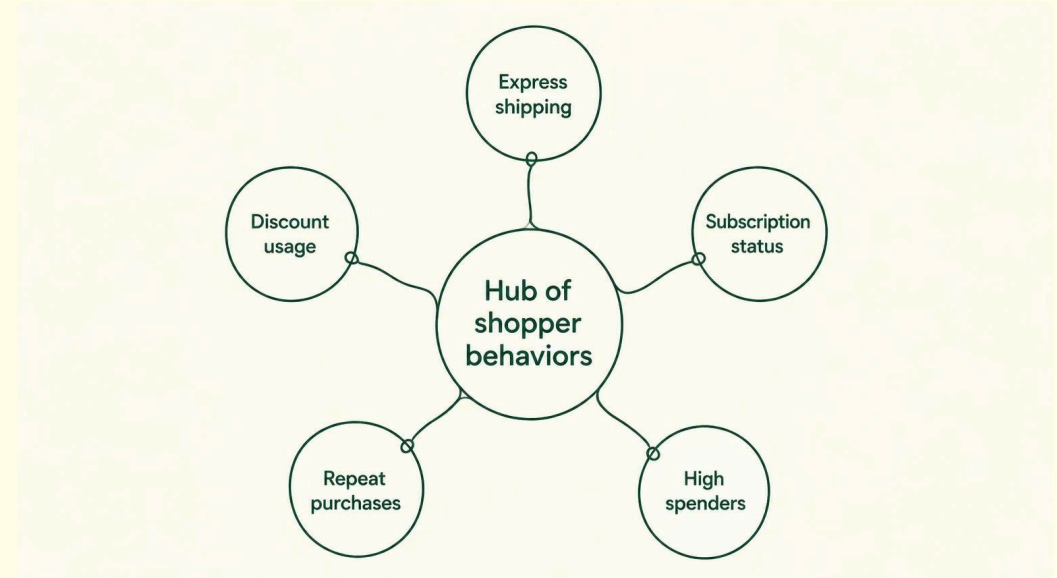
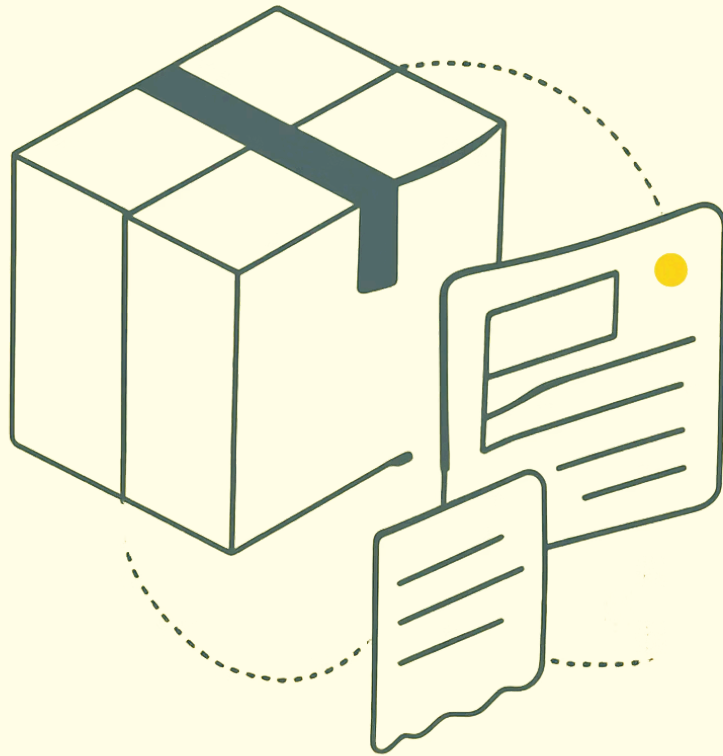


## Discount-Dependent Products

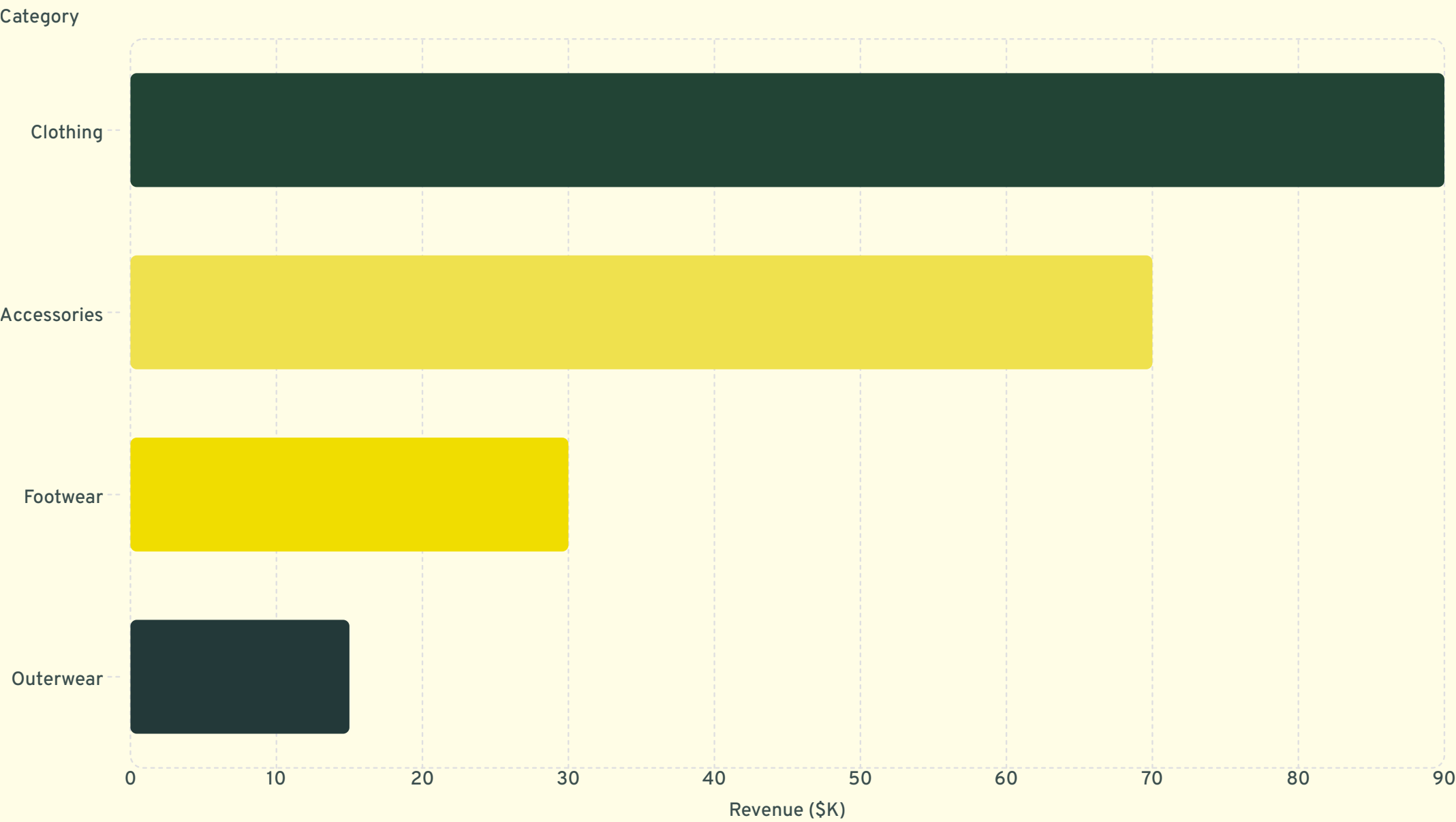
Highest discount rates: Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37 – review discount policy for margin impact.

# Behavioral Insights – Spending & Shipping

Shipping comparison: Express orders average \$60.48 vs Standard \$58.46. High-spending discount users (total rows: 839) were identified – customers who used discounts yet spent above average, indicating profitable promotional responsiveness for certain segments.



# Revenue & Sales by Category and Age Group



Revenue concentration: Clothing (~\$90K) and Accessories (~\$70K) dominate. Revenue by age group: Young Adult ~\$62K, Middle-aged ~\$59K, Adult ~\$56K, Senior ~\$56K – prioritize campaigns by revenue contribution and lifecycle stage.

# Power BI Dashboard Snapshot

We built an interactive Power BI dashboard to visualize key metrics: 3.9K customers, \$59.76 average purchase, 3.75 average rating, subscription split (Yes 27% / No 73%). Visuals include revenue & sales by category, revenue and sales by age group, and subscription filters for targeted exploration.





# Business Recommendations & Next Steps

## Boost Subscriptions

Promote exclusive subscriber benefits to grow recurring revenue and retention among high-frequency buyers.

## Loyalty Programs

Reward repeat buyers to shift customers into the Loyal segment and increase lifetime value.

## Review Discount Policy

Balance promotional volume with margins; target discounts to products that drive profitable incremental sales.

## Product Positioning & Targeting

Highlight top-rated and top-selling items in campaigns and focus marketing on high-revenue age groups and express-shipping customers.

- Next steps: A/B test subscription incentives, refine discount segmentation, and expand Power BI dashboards with cohort and CLTV analysis.